

(*) On every Riemannian manifold, we will always use the Levi-Civita connection if there is no further comment.

(*) Throughout this note, we always assume that $(\widetilde{M}, \widetilde{g})$ is an m -dimensional Riemannian manifold and (M, g) is an embedded n -dimensional Riemannian submanifold of $\widetilde{M}$. Here $g = \iota^* \widetilde{g}$ is the induced metric on M , with $\iota : M \hookrightarrow \widetilde{M}$.

1 Second Fundamental Form

1.1 Gauss Formula

Recall that for an embedded submanifold $M \subseteq \mathbb{R}^n$, we have defined the tangential bundle TM and the normal bundle NM on M , where $TM = \cup_{p \in M} T_p M$, $NM = \cup_{p \in M} N_p M$ and $T_p \mathbb{R}^n = T_p M \oplus N_p M$ for any $p \in M$. Moreover, we have the bundle homomorphisms

$$\pi^\top : T\mathbb{R}^n|_M \rightarrow TM$$

$$\pi^\perp : T\mathbb{R}^n|_M \rightarrow NM$$

which are just the projections. Now we do the same thing with the embedded submanifold $M \subseteq \widetilde{M}$, let

$$\pi^\top : T\widetilde{M}|_M \rightarrow TM$$

$$\pi^\perp : T\widetilde{M}|_M \rightarrow NM$$

be the projections, they are called **tangential projection** and **normal projection** respectively. For $\forall X \in \Gamma(T\widetilde{M}|_M)$, denote $X^\top := \pi^\top(X)$ and $X^\perp := \pi^\perp(X)$.

Let $\widetilde{\nabla}$ be the Levi-Civita connection of $\widetilde{M}$ and ∇ be that of M , for any two vector fields $X, Y \in \mathfrak{X}(M) = \Gamma(TM)$, we can extend them to an open subset of $\widetilde{M}$ (still denote by X, Y), then we have the decomposition

$$\widetilde{\nabla}_X Y = (\widetilde{\nabla}_X Y)^\top + (\widetilde{\nabla}_X Y)^\perp$$

Second Fundamental Form

Definition 1.1. : Let the assumption be as the above discussion, we define the **second fundamental form** to be the map

$$\Pi : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \Gamma(NM)$$

$$(X, Y) \mapsto \Pi(X, Y) := (\widetilde{\nabla}_X Y)^\perp$$

where X, Y are extended arbitrarily to an open subset of $\widetilde{M}$.

Remark: The **first fundamental form** was used to refer to the induced metric $g = \iota^* \widetilde{g}$ on M , here $\iota : M \hookrightarrow \widetilde{M}$. The word *form* in both cases refers to bilinear forms, not differential forms!

Properties of Second Fundamental Form

Proposition 1.1. : Let M be an embedded submanifold of $(\widetilde{M}, \widetilde{g})$, then for $X, Y \in \mathfrak{X}(M)$, we have

- (1) $\Pi(X, Y)$ is independent of the extension of X and Y to an open subset of $\widetilde{M}$;
- (2) $\Pi(X, Y)$ is bilinear over $C^\infty(M)$ in X and Y ;
- (3) $\Pi(X, Y)$ is symmetric in X and Y ;
- (4) The value of $\Pi(X, Y)$ at a point $p \in M$ depends only on X_p and Y_p .

Proof: We first prove (3): choose a particular extension of X and Y to a neighborhood of M in $\widetilde{M}$, by the torsion-freeness of $\widetilde{\nabla}$, we get

$$\Pi(X, Y) - \Pi(Y, X) = (\widetilde{\nabla}_X Y - \widetilde{\nabla}_Y X)^\perp = [X, Y]^\perp$$

note that $[X, Y] \in \mathfrak{X}(M)$ since $X, Y \in \mathfrak{X}(M)$, which means that $[X, Y]^\perp = 0$;

Now we prove (1) and (4), just only need to recall that by the property of connections on TM , $\widetilde{\nabla}_X Y|_p$ depends only on X_p once we fix Y , so the same is true for $\Pi(X, Y)$, the same argument applies to Y as well.

Now we prove (2), by definition, we know that $\widetilde{\nabla}_X Y$ is $C^\infty(M)$ -linear over X . By the theory of smooth manifolds, we know that for every $f \in C^\infty(M)$, it can be extended to a smooth function on a neighborhood of M in $\widetilde{M}$, thus $\Pi(X, Y)$ is linear over $C^\infty(M)$ in X , by the symmetry of it, the same claim holds for Y . $\square$

Remark: By this proposition, for $v, w \in T_p M$, it makes sense to interpret $\Pi(V, W)|_p = \Pi(v, w)$ as long as the vector fields $V, W \in \mathfrak{X}(M)$ satisfy $V_p = v, W_p = w$.

Gauss Formula

Theorem 1.1. : Let (M, g) be an embedded submanifold of $(\widetilde{M}, \widetilde{g})$. If $X, Y \in \mathfrak{X}(M)$ are extended arbitrarily to smooth vector fields on a neighborhood of M in $\widetilde{M}$, then we have

$$\widetilde{\nabla}_X Y = \nabla_X Y + \Pi(X, Y).$$

Proof: Since $\widetilde{\nabla}_X Y = (\widetilde{\nabla}_X Y)^\top + (\widetilde{\nabla}_X Y)^\perp$, we only need to show that $\nabla_X Y = (\widetilde{\nabla}_X Y)^\top$.

Now, we define a map

$$\begin{aligned} \nabla^\top : \mathfrak{X}(M) \times \mathfrak{X}(M) &\rightarrow \mathfrak{X}(M) \\ (X, Y) &\mapsto \nabla_X^\top Y := (\widetilde{\nabla}_X Y)^\top \end{aligned}$$

where X, Y are extended arbitrarily to an open subset of $\widetilde{M}$ containing M . First we show that $\nabla^\top$ is a connection. It's clear that $\nabla_X^\top Y$ is $C^\infty(M)$ -linear in X and $\mathbb{R}$ -linear in Y by the definition of it, we just need to check the Leibniz rule. Take $\forall f \in C^\infty(M)$, let $\widetilde{f}$ be an extension of f to the open subset of $\widetilde{M}$ containing M , thus $\widetilde{f}Y$ is a smooth extension of fY to the open subset, hence we get

$$\nabla_X^\top(fY) = (\widetilde{\nabla}_X fY)^\top = \pi^\top(\widetilde{\nabla}_X fY) = \pi^\top(X(f)Y + f\widetilde{\nabla}_X Y) = \pi^\top(X(f)Y) + \pi^\top(f\widetilde{\nabla}_X Y)$$

note that $X(f)Y \in \mathfrak{X}(M)$ hence $\pi^\top(X(f)Y) = X(f)Y$ and that $\pi^\top$ is a bundle homomorphism hence $\pi^\top(f\tilde{\nabla}_X Y) = f\pi^\top(\tilde{\nabla}_X Y) = f\nabla_X^\top Y$, i.e.,

$$\nabla_X^\top(fY) = X(f)Y + f\nabla_X^\top Y$$

which is exactly the product rule. Thus $\nabla^\top$ is indeed a connection. Second, we show that $\nabla^\top$ is compatible with the metric. Take $\forall X, Y, Z \in \mathfrak{X}(M)$, let $\tilde{X}, \tilde{Y}, \tilde{Z}$ be smooth extensions of them respectively on an open subset of $\tilde{M}$ containing M , so on any points of M we get

$$\nabla_X^\top \langle Y, Z \rangle = X \langle Y, Z \rangle = \tilde{X} \langle \tilde{Y}, \tilde{Z} \rangle = \tilde{\nabla}_{\tilde{X}} \langle \tilde{Y}, \tilde{Z} \rangle = \langle \tilde{\nabla}_{\tilde{X}} \tilde{Y}, \tilde{Z} \rangle + \langle \tilde{Y}, \tilde{\nabla}_{\tilde{X}} \tilde{Z} \rangle$$

note that on any points of M , we have $\langle \tilde{\nabla}_{\tilde{X}} \tilde{Y}, \tilde{Z} \rangle = \langle \pi^\top(\tilde{\nabla}_{\tilde{X}} \tilde{Y}), \tilde{Z} \rangle$ and $\langle \tilde{Y}, \tilde{\nabla}_{\tilde{X}} \tilde{Z} \rangle = \langle \tilde{Y}, \pi^\top(\tilde{\nabla}_{\tilde{X}} \tilde{Z}) \rangle$, which means that on any points of M

$$\nabla_X^\top \langle Y, Z \rangle = \langle \tilde{\nabla}_{\tilde{X}} \tilde{Y}, \tilde{Z} \rangle + \langle \tilde{Y}, \tilde{\nabla}_{\tilde{X}} \tilde{Z} \rangle = \langle \pi^\top(\tilde{\nabla}_{\tilde{X}} \tilde{Y}), \tilde{Z} \rangle + \langle \tilde{Y}, \pi^\top(\tilde{\nabla}_{\tilde{X}} \tilde{Z}) \rangle = \langle \nabla_X^\top Y, Z \rangle + \langle Y, \nabla_X^\top Z \rangle$$

hence $\nabla^\top$ is compatible with the metric g . Finally we show that $\nabla^\top$ is torison free (or we say symmetric). Let $\iota : M \rightarrow \tilde{M}$ denote the inclusion map, and $X, Y, \tilde{X}, \tilde{Y}$ be as above, then we know that X, Y are ι -related to $\tilde{X}, \tilde{Y}$ respectively, hence by the theory of smooth manifolds, we can know that $[X, Y]$ is also ι -related to $[\tilde{X}, \tilde{Y}]$. More ove, we have $[\tilde{X}, \tilde{Y}]|_M = [X, Y] \in \mathfrak{X}(M)$, thus

$$\nabla_X^\top Y - \nabla_Y^\top X = \pi^\top(\tilde{\nabla}_{\tilde{X}} \tilde{Y}|_M - \tilde{\nabla}_{\tilde{Y}} \tilde{X}|_M) = \pi^\top([\tilde{X}, \tilde{Y}]|_M) = [\tilde{X}, \tilde{Y}]|_M = [X, Y]$$

so $\nabla^\top$ is torison free (or symmetric).

By the discussion above, we know that $\nabla^\top$ is the Levi-Civita connection on M , by the uniqueness of Levi-Civita connection, we get $\nabla^\top = \nabla$. Since by definition $\Pi(X, Y) = (\tilde{\nabla}_X Y)^\perp$, the decomposition rewrites as

$$\tilde{\nabla}_X Y = \nabla_X Y + \Pi(X, Y)$$

which is the conclusion. $\square$

For the Riemannian submanifold $M \subseteq \tilde{M}$, let $D_t X$ and $\tilde{D}_t X$ denote the covariant derivatives of $X \in \mathfrak{X}(\gamma)$ along γ as a curve in M and as a curve in $\tilde{M}$ respectively, where $\gamma : I \rightarrow M$ is a smooth curve.

Gauss Formula along Curves

Corollary 1.1. : *Let (M, g) be an embedded submanifold of $(\tilde{M}, \tilde{g})$ and $\gamma : I \rightarrow M$ is a smooth curve. If X is a smooth curve along γ that is everywhere tangent to M , then*

$$\tilde{D}_t X = D_t X + \Pi(\gamma', X).$$

Proof: For any $t_0 \in I$, by the theory of smooth manifolds, there exists an adapted orthonormal frame $(E_1, E_2, \dots, E_m)$ in a neighborhood of $\gamma(t_0)$, i.e., for any point p in such neighborhood, $(E_1|_p, E_2|_p, \dots, E_n|_p)$ is an orthonormal frame of $T_p M$ and $(E_1|_p, E_2|_p, \dots, E_m|_p)$ is an orthonormal frame of $T_p \tilde{M}$. So in terms of such frames, we can write

$$X(t) \triangleq \sum_{i=1}^n X^i(t) E_i|_{\gamma(t)}$$

by the product rule and the Gauss formula, we get

$$\begin{aligned}\tilde{D}_t X &= \sum_{i=1}^n (\dot{X}^i E_i + X^i \tilde{\nabla}_{\gamma'} E_i) \\ &= \sum_{i=1}^n (\dot{X}^i E_i + X^i \nabla_{\gamma'} E_i + X^i \Pi(\gamma', E_i)) = D_t X + \Pi(\gamma', X)\end{aligned}$$

where the last equation is by the bilinearity of Π . $\square$

1.2 Gauss Equation

For each normal vector field $N \in \Gamma(NM)$, we obtain a scalar-valued symmetric bilinear form $\Pi_N : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow C^\infty(M)$, which is given by

$$\Pi_N(X, Y) := \langle N, \Pi(X, Y) \rangle$$

Weingarten Map

Definition 1.2. : Let $N \in \Gamma(NM)$, define the **Weingarten map in the direction of N** to be the self-adjoint linear map $W_N : \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$, which is characterized by the equation

$$\langle W_N(X), Y \rangle = \Pi_N(X, Y) = \langle N, \Pi(X, Y) \rangle$$

Remark: Since the second fundamental form Π is $C^\infty(M)$ -bilinear, it can be seen that W_N is $C^\infty(M)$ -linear, thus W_N defines a bundle homomorphism from TM to itself by the theory of bundle homomorphisms.

Weingarten Equation

Proposition 1.2. : Let (M, g) be an embedded submanifold of $(\tilde{M}, \tilde{g})$, then for any $X \in \mathfrak{X}(M)$ and any $N \in \Gamma(NM)$, we have the following equation

$$(\tilde{\nabla}_X N)^\top = -W_N(X)$$

where N is extended arbitrarily to an open subset of $\tilde{M}$ containing M .

Proof: For any $Y \in \mathfrak{X}(M)$, extend Y to a vector field on the open subset of $\tilde{M}$ containing M . Thus on any points of M , we have

$$\begin{aligned}0 &\equiv X \langle N, Y \rangle = \langle \tilde{\nabla}_X N, Y \rangle + \langle N, \tilde{\nabla}_X Y \rangle = \langle \tilde{\nabla}_X N, Y \rangle + \langle N, \nabla_X Y + \Pi(X, Y) \rangle \\ &= \langle \tilde{\nabla}_X N, Y \rangle + \langle N, \Pi(X, Y) \rangle = \langle \tilde{\nabla}_X N, Y \rangle + \langle W_N(X), Y \rangle.\end{aligned}$$

Since Y is taken arbitrarily, hence we get

$$(\tilde{\nabla}_X N + W_N(X))^\top = (\tilde{\nabla}_X N)^\top + W_N(X) = 0$$

which is the conclusion. $\square$

The second fundamental form plays an important role in describing the difference between the curvature tensors of $\widetilde{M}$ and M .

Gauss Equation

Theorem 1.2. : Let (M, g) be an embedded submanifold of $(\widetilde{M}, \widetilde{g})$, for any $W, X, Y, Z \in \mathfrak{X}(M)$, we have the following equation

$$\widetilde{Rm}(W, X, Y, Z) = Rm(W, X, Y, Z) - \langle \Pi(W, Z), \Pi(X, Y) \rangle + \langle \Pi(W, Y), \Pi(X, Z) \rangle$$

Proof: Extend W, X, Y, Z arbitrarily to an open subset of $\widetilde{M}$ containing M . By the definition of curvature and the Gauss Formula, at any points of M , we have

$$\begin{aligned} \widetilde{Rm}(W, X, Y, Z) &= \langle \widetilde{\nabla}_W \widetilde{\nabla}_X Y - \widetilde{\nabla}_X \widetilde{\nabla}_W Y - \widetilde{\nabla}_{[W, X]} Y, Z \rangle \\ &= \langle \widetilde{\nabla}_W (\nabla_X Y + \Pi(X, Y)) - \widetilde{\nabla}_X (\nabla_W Y + \Pi(W, Y)) - \widetilde{\nabla}_{[W, X]} Y, Z \rangle \end{aligned}$$

note that $\langle \widetilde{\nabla}_W \Pi(X, Y), Z \rangle = \langle (\widetilde{\nabla}_W \Pi(X, Y))^\top, Z \rangle$ since $Z \in \mathfrak{X}(M)$, by the Weingarten equation, taking $N = \Pi(X, Y)$ and $N = \Pi(W, Y)$, we get

$$\widetilde{Rm}(W, X, Y, Z) = \langle \widetilde{\nabla}_W \nabla_X Y, Z \rangle - \langle \widetilde{\nabla}_X \nabla_W Y, Z \rangle - \langle \widetilde{\nabla}_{[W, X]} Y, Z \rangle + \langle -W_{\Pi(X, Y)}(W), Z \rangle - \langle -W_{\Pi(W, Y)}(X), Z \rangle$$

hence by the definition of the Weingarten map, we get

$$\widetilde{Rm}(W, X, Y, Z) = \langle \widetilde{\nabla}_W \nabla_X Y, Z \rangle - \langle \widetilde{\nabla}_X \nabla_W Y, Z \rangle - \langle \widetilde{\nabla}_{[W, X]} Y, Z \rangle - \langle \Pi(X, Y), \Pi(W, Z) \rangle + \langle \Pi(W, Y), \Pi(X, Z) \rangle$$

by Gauss formula again, it can be seen that only the tangential components survive, i.e.,

$$\begin{aligned} \widetilde{Rm}(W, X, Y, Z) &= \langle \nabla_W \nabla_X Y, Z \rangle - \langle \nabla_X \nabla_W Y, Z \rangle - \langle \nabla_{[W, X]} Y, Z \rangle - \langle \Pi(X, Y), \Pi(W, Z) \rangle + \langle \Pi(W, Y), \Pi(X, Z) \rangle \\ &= \langle R(W, X)Y, Z \rangle - \langle \Pi(X, Y), \Pi(W, Z) \rangle + \langle \Pi(W, Y), \Pi(X, Z) \rangle \end{aligned}$$

which is the desired conclusion. $\square$

1.3 Codazzi Equation

Let (M, g) be an embedded submanifold of $(\widetilde{M}, \widetilde{g})$, we define the **normal connection** as

$$\nabla^\perp : \mathfrak{X}(M) \times \Gamma(NM) \rightarrow \Gamma(NM)$$

$$(X, N) \mapsto \nabla_X^\perp N := (\widetilde{\nabla}_X N)^\perp$$

where N is extended to a smooth vector field on an open subset of $\widetilde{M}$ containing M . Just as in the proof of the Gauss formula, we can check that such defined $\nabla^\perp$ is indeed a connection on the normal bundle NM , which is also compatible with the metric, i.e., $\nabla_X^\perp \langle N_1, N_2 \rangle = \langle \nabla_X^\perp N_1, N_2 \rangle + \langle N_1, \nabla_X^\perp N_2 \rangle$.

Now, consider a vector bundle $F \rightarrow M$ whose fiber at each point $p \in M$ is $F_p := \{f \mid f : T_p M \times T_p M \rightarrow N_p M \text{ is bilinear}\}$, hence for any $\sigma \in \Gamma(F)$, we have $\sigma(p) \in F_p$. So the smooth sections of F correspond to smooth maps $\mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \Gamma(NM)$ that are $C^\infty(M)$ -bilinear, which implies that $\Pi \in \Gamma(F)$. We define the connection ∇^F as follows:

$$\nabla^F : \mathfrak{X}(M) \times \Gamma(F) \rightarrow \Gamma(F)$$

$$(X, B) \mapsto \nabla_X^F B$$

where $(\nabla_X^F B)(Y, Z) := \nabla_X^\perp(B(Y, Z)) - B(\nabla_X Y, Z) - B(Y, \nabla_X Z)$.

Codazzi Equation

Theorem 1.3. : Let (M, g) be an embedded submanifold of $(\widetilde{M}, \widetilde{g})$, for $\forall W, X, Y \in \mathfrak{X}(M)$, we have the following equation

$$(\widetilde{R}(W, X)Y)^\perp = (\nabla_W^F \Pi)(X, Y) - (\nabla_X^F \Pi)(W, Y)$$

Proof: For any $N \in \Gamma(NM)$, we have

$$\langle \widetilde{R}(W, X)Y, N \rangle = \langle (\widetilde{R}(W, X)Y)^\perp, N \rangle$$

Extend W, X, Y arbitrarily to an open subset of $\widetilde{M}$ containing M , by the Gauss formula, we get

$$\widetilde{Rm}(W, X, Y, N) = \langle \widetilde{R}(W, X)Y, N \rangle = \langle \widetilde{\nabla}_W(\nabla_X Y + \Pi(X, Y)) - \widetilde{\nabla}_X(\nabla_W Y + \Pi(W, Y)) - \widetilde{\nabla}_{[W, X]}Y, N \rangle$$

since we only need to consider the normal components, by the definition of the normal connection $\nabla^\perp$ and the connection ∇^F , by Gauss formula again, we get

$$\begin{aligned} \widetilde{Rm}(W, X, Y, N) &= \langle \Pi(W, \nabla_X Y) + (\nabla_W^F \Pi)(X, Y) + \Pi(\nabla_W X, Y) + \Pi(X, \nabla_W Y), N \rangle \\ &\quad - \langle \Pi(X, \nabla_W Y) + (\nabla_X^F \Pi)(W, Y) + \Pi(\nabla_X W, Y) + \Pi(W, \nabla_X Y), N \rangle - \langle \Pi([W, X], Y), N \rangle \end{aligned}$$

just note that ∇ is torsion-free, i.e., $\nabla_W X - \nabla_X W = [W, X]$. Hence we get

$$\widetilde{Rm}(W, X, Y, N) = \langle \widetilde{R}(W, X)Y, N \rangle = \langle (\widetilde{R}(W, X)Y)^\perp, N \rangle = \langle (\nabla_W^F \Pi)(X, Y) - (\nabla_X^F \Pi)(W, Y), N \rangle$$

which implies the conclusion. $\square$

2 Curvature of Curves

Recall that for a smooth curve $\gamma : I \rightarrow M$, we have defined its **acceleration vector field** to be $D_t \gamma'$, where D_t is the covariant derivative along γ .

Geodesic Curvature of a Curve

Definition 2.1. : Let (M, g) be a Riemannian manifold, $\gamma : I \rightarrow M$ is a smooth unit-speed curve in M . (Recall that any smooth curve in M has a unit-speed reparametrization) We define the **(geodesic) curvature of γ** as the length of the acceleration vector field, which is a function $\kappa : I \rightarrow \mathbb{R}$ given by

$$\kappa(t) := |D_t \gamma'(t)|$$

Remark: For such a curve $\gamma : I \rightarrow M$, it has two distinct geodesic curvatures: the **intrinsic curvature** κ as a curve in M and **extrinsic curvature** $\widetilde{\kappa}$ as a curve in $\widetilde{M}$. i.e., $\kappa(t) = |D_t \gamma'(t)|$ and $\widetilde{\kappa}(t) = |\widetilde{D}_t \gamma'(t)|$.

Geometric Interpretation of Π

Proposition 2.1. : Let (M, g) be an embedded submanifold of $(\widetilde{M}, \widetilde{g})$, for $p \in M$ and $v \in T_p M$, we have

- (1) $\Pi(v, v)$ is the $\widetilde{g}$ -geodesic acceleration at p of the g -geodesic γ_v , i.e., $\widetilde{D}_t \gamma'_v|_p = \Pi(v, v)$.
- (2) If v is a unit vector, then $|\Pi(v, v)|$ is the $\widetilde{g}$ -curvature of the g -geodesic γ_v at p .

Proof: (1) For the given $p \in M$ and $v \in T_p M$, choose a regular curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow M$ such that $\gamma(0) = p$ and $\gamma'(0) = v$. By the Gauss formula along curves, we get

$$\widetilde{D}_t \gamma' = D_t \gamma' + \Pi(\gamma', \gamma')$$

so if $\gamma = \gamma_v$ is the g -geodesic in M , we have $D_t \gamma' = 0$, which means that

$$\widetilde{D}_t \gamma' = \Pi(\gamma', \gamma')$$

evaluate at $t = 0$, we get $\widetilde{D}_t \gamma'|_p = \Pi(v, v)$.

(2) By the definition of geodesic curvature. $\square$

For a smooth curve $\gamma : I \rightarrow M$, its intrinsic and extrinsic acceleration of it are usually different, it is generally not the case that a $\widetilde{g}$ -geodesic that starts tangent to M stays in M . A Riemannian submanifold (M, g) of $(\widetilde{M}, \widetilde{g})$ is said to be **totally geodeisc** if every $\widetilde{g}$ -geodesic that is tangent to M at some time t_0 stays in M for $\forall t \in (t_0 - \varepsilon, t_0 + \varepsilon)$.

Proposition 2.2. : Let (M, g) be an embedded submanifold of $(\widetilde{M}, \widetilde{g})$, then TFAE:

- (1) M is totally geodesic in $\widetilde{M}$;
- (2) Every g -geodesic in M is also a $\widetilde{g}$ -geodesic in $\widetilde{M}$;
- (3) The second fundamental form of M vanishes identically.

Proof: (1) $\Rightarrow$ (2): Let $\gamma : I \rightarrow M$ be a g -geodesic in M . For $t_0 \in I$, let $\widetilde{\gamma} : \widetilde{I} \rightarrow \widetilde{M}$ be the $\widetilde{g}$ -geodesic such that $\widetilde{\gamma}(t_0) = \gamma(t_0)$ and $\widetilde{\gamma}'(t_0) = \gamma'(t_0)$ (by the theory of geodesics we know that such $\widetilde{\gamma}$ exists). So by the given condition, there is some open interval I_0 such that $t_0 \in I_0$ and $\widetilde{\gamma}(t)|_{I_0} \in M$. By the Gauss formula along curves, on the interval I_0 , we get

$$\widetilde{D}_t \widetilde{\gamma}' = 0 = D_t \widetilde{\gamma}' + \Pi(\widetilde{\gamma}', \widetilde{\gamma}')$$

note that $D_t \widetilde{\gamma}' \in \mathfrak{X}(M)$ and $\Pi(\widetilde{\gamma}', \widetilde{\gamma}') \in \Gamma(NM)$, so it must be $D_t \widetilde{\gamma}' = \Pi(\widetilde{\gamma}', \widetilde{\gamma}') = 0$. In particular, $D_t \widetilde{\gamma}'|_{I_0} \equiv 0$, which means that $\widetilde{\gamma}$ is also a g -geodesic on the interval I_0 . So by the uniqueness of geodesics, we get $\widetilde{\gamma}|_{I_0} = \gamma|_{I_0}$, so we know that γ is also a $\widetilde{g}$ -geodeisc. Note that the time t_0 is taken arbitrarily, thus γ is a $\widetilde{g}$ -geodeisc in the whole domain.

(2) $\Rightarrow$ (3): For $\forall p \in M$ and $\forall v \in T_p M$, let $\gamma = \gamma_v : I \rightarrow M$ be the g -geodeisc such that $\gamma(0) = p$ and $\gamma'(0) = v$. By the condition, we know that γ is also a $\widetilde{g}$ -geodesic, hence we have

$$D_t \gamma' = \widetilde{D}_t \gamma' = 0$$

so by the Gauss formula along curves, it can be seen that $\Pi(\gamma', \gamma') = 0$, evaluate at $t = 0$, we obtain $\Pi(v, v) = 0$, note that v is taken arbitrarily, so by the theory of linear algebra, we get $\Pi = 0$;

(3) $\Rightarrow$ (1): Let $\tilde{\gamma} : I \rightarrow \tilde{M}$ be a $\tilde{g}$ -geodeisc such that $\tilde{\gamma}(t_0) \in M$ for some time $t_0 \in I$. Let $\gamma : I \rightarrow M$ be the g -geodeisc with $\gamma(t_0) = \tilde{\gamma}(t_0)$ and $\gamma'(t_0) = \tilde{\gamma}'(t_0)$. By the condition that $\Pi \equiv 0$ and the Gauss formula, we get

$$\tilde{D}_t \gamma' = D_t \gamma' = 0$$

which means that γ is also a $\tilde{g}$ -geodeisc. By the uniqueness of geodesics, we know that $\gamma = \tilde{\gamma}$ on their common domain, this implies that $\tilde{\gamma}(t)$ lies in M for some open interval containing t_0 . $\square$

Remark: In the proof in (2) $\Rightarrow$ (3), we have used the fact in linear algebra that for an inner product space V and a vector space W , if $B, B' : V \times V \rightarrow W$ are bilinear and symmetric and for any vector $v \in V$, we have $B(v, v) = B'(v, v)$, then $B = B'$. To see the proof of this, just consider the identity that $B(v, w) = \frac{1}{4}(B(v+w, v+w) - B(v-w, v-w))$ for any $v, w \in V$. So in the case of second fundamental form, this means that Π is completely determined by its value of the form $\Pi(v, v)$ for $v \in TM$.

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